DeepDT: Generative Adversarial Network for High-Resolution Climate Prediction

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***Abstract*— Climate prediction is susceptible to a variety of meteorological factors, and downscaling technology is used for high-resolution climate prediction. This technology can generate small-scale regional climate prediction from large-scale climate output information. Inspired by the concept of image super resolution, we propose to apply the convolutional neural network (CNN) to downscaling technology. However, some unpleasant artifacts always appear in the final climate images generated by existing CNN-based models. To further eliminate these unpleasant artifacts, we present a new training strategy for the generative adversarial network, termed DeepDT. The key idea of our DeepDT is to train a generator and a discriminator separately. More specifically, we apply the residual-in-residual dense block as the basic frame structure to fully extract the features of the input. Additionally, we innovatively use a CNN model to fuse multiple climate elements to generate trainable climate images, and build a high-quality climate data set. Finally, we evaluate the DeepDT using the proposed climate data sets, and the experiments indicate that DeepDT performs best compared to most CNN-based models in climate prediction.**

***Index Terms*— Climate prediction, generative adversarial network, image super resolution.**

I. Introduction

HIGH-RESOLUTION climate prediction [1] has always been a hard task, and its appearance is susceptible to a variety of meteorological factors. We get small-scale regional climate forecasts from large-scale meteorological record information, referred to as high-resolution (or high-quality) climate prediction [2]. For example, we can produce 1-km (high-resolution) climate forecast from the low-resolution climate observation data at a 10-km small scale, as shown in Fig. 1. Additionally, climate prediction contains many weather patterns, and we only discuss the precipitation prediction in this letter. Note that some meteorological factors have an impact on the performance of precipitation forecasts, such as topography, temperature, and humidity.

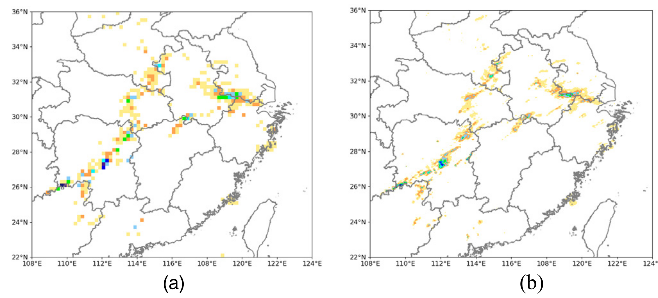

Fig. 1. Low-resolution and high-resolution precipitation data. (a) 10-km low resolution. (b) 1-km high resolution.

Downscaling technology is always used for climate prediction, and traditional downscaling methods mainly used the dynamical and statistical models, such as linear models [4]. However, these early algorithms just learned a function from the observed climate data, and did not consider the influences of other climate factors.

The purpose of image super resolution [5] is to output a high-quality image from its poor visual input image. Motivated by this concept, we propose to apply the super-resolution models to downscaling technology. For example, Dong *et al.* [6] and Lai *et al.* [7] first presented the convolutional neural network (CNN) to extract nonlinear features from the input images. This model, termed SRCNN, outperformed traditional methods. Later, Zhang *et al.* [8] applied the Laplacian pyramid to the image super resolution (LapSRN). This model simultaneously generated high-resolution images of different scales, which reduced the difficulty of model training. Ledig *et al.* [9] applied the residual dense block for image super resolution (RDN). This method used a novel residual learning method to extract feature maps from the original input images. However, these algorithms were mainly optimized by minimizing the mean squared error (MSE), which resulted in lacking some high-frequency content in the final generated images.

To solve these drawbacks, Odena *et al.* [10] introduced an adversarial neural network in image super resolution (SRGAN). The general idea of SRGAN was that it allowed one to train a generator to deceive a discriminator that was trained to discriminate high-resolution output results from the label image. Additionally, SRGAN defined a novel perceptual loss [11], which differed from traditional loss functions such

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as MSE. Unfortunately, some unpleasant artifacts [12] still accompanied the final generated images. Therefore, we borrow some ideas from the enhanced super-resolution generative adversarial network (ESRGAN) [13] and propose the DeepDT method. Our model uses a more efficient residual block to learn features from all layers, not the original basic block in SRGAN. Also, we apply a simplified perceptual loss to optimize our network. Additionally, Sheikholeslami *et al.* [14] presented the efficient unsupervised super-resolution model (EUSR) for remote images. EUSR applied dense skip connections and improved image reconstruction performance. However, the China meteorological administration (CMA) has provided us with low-resolution and high-resolution data pairs, and the supervised learning model can perform better than the unsupervised super-resolution model. Finally, we innovatively apply CNN to merge multiple meteorological factors into climate images, which are used to build the NCNI high-quality climate image data set.

The main contributions of this letter are summarized as follows.

1) We apply the CNN-based models to downscaling technology, and innovatively propose the DeepDT model. DeepDT is an efficient generative adversarial network for high-resolution climate prediction.
2) We propose to use CNN to fuse multiple related climate factors to generate trainable climate images. This measure considers the influences of different climate factors on a certain climate forecast.
3) We construct a high-quality climate image data set (NCNI) and evaluate all models on this benchmark data set. If researchers need these climate data for scientific research, you can contact the corresponding author by email.

II. Methodology

The proposed DeepDT consists of three parts. The first part is to introduce the process of image preprocessing. We combine raw climate data with different climate elements to generate trainable climate images. The second part is to introduce our downscaling method. Our algorithm makes further adjustments on SRGAN. The third part is to introduce a simplified perceptual loss function.

A. CNN for Image Preprocessing

As mentioned above, the quality of precipitation prediction is affected by topography, temperature, and humidity climate variables. Besides, the general colored image includes three channels such as red, blue, and green. Each channel represents a feature of the picture. Inspired by this structure, we fill meteorological element data in each image channel and get the multichannel climate images for model input. Furthermore, our original climate data are only available at the CMA. The following describes the generation process of the model input image.

First, we get all relevant meteorological data from CMA, such as 3-D low-resolution precipitation data I_{low}, 2-D topography data I_{to}, 3-D temperature data I_{tem}, and 3-D humidity data I_{hum}. Then, we expand the low-dimensional topography data I_{to} $(N \times M)$ into 3-D data I_{top} $(N \times M \times 1)$, so that all the meteorological data have the same dimension for image fusion. Next, we regard each climate factors as a channel and attain the 4-D climate images I_{ori}.

Finally, we pass the 4-D images I_{ori} (four-channel) through a 1×1 convolutional layer with the number of filters is three, and get the trainable climate images I_{LR} (three-channel), as shown in Fig. 2. By doing so, we can reduce the dimensions of the input climate image and greatly increase the nonlinear characteristics of the climate image. Furthermore, we use these images to construct the NCNI data set, which is a high-quality climate data set for climate forecast.

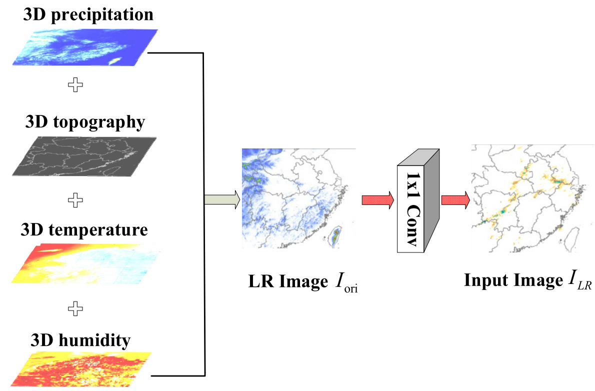

Fig. 2. Steps to generate model input climate images. The original low-resolution image has four channels, the final input image has three channels, and the final output images have only one channel.

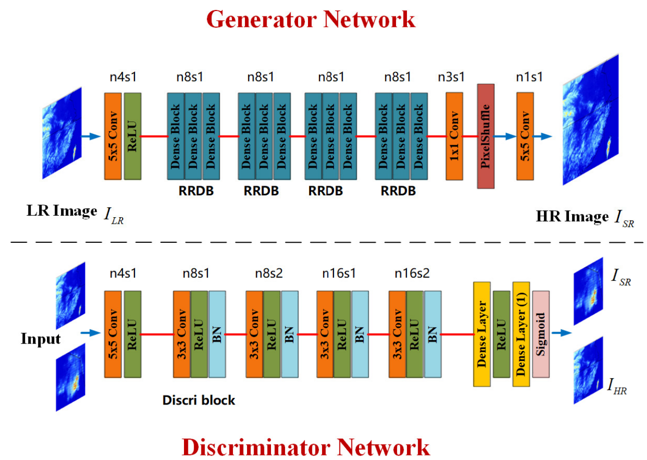

Fig. 3. General framework of the DeepDT. n refers to the feature maps of each Conv layer, and s refers to the stride.

B. Network Structure

Our DeepDT mainly includes two structures: generator network and discriminator network, as shown in Fig. 3.

1) Generator Network: We use this network to estimate its corresponding high-resolution counterpart I_{SR} for a given low-resolution input image I_{LR}. The generator has two 5×5 Conv layers and four consecutive residual-in-residual dense blocks (RRDB). Note that RRDB is applied as the basic frame structure and contains three dense blocks [15], as shown in Fig. 4. Among them, each block contains three 3×3 Conv layers, one 1×1 Conv layer, and three ReLU functions.

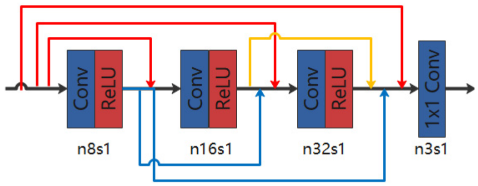

Fig. 4. Internal framework of a dense block. n refers to the feature maps of each Conv layer, and s refers to the stride. The last 1×1 Conv layer has three filter maps.

Specially, we use a 1×1 Conv layer the reduce the dimensions of the RRDB output data.

Compared with the basic block in SRGAN, our RRDB uses the dense residual learning strategy, which can improve the representation ability of extracted features. In particular, we observe that the BN layer is easy to bring unpleasant artifacts when the GAN-based model is deeper. Thereby, we remove all BN layers in the generator. Furthermore, we apply the pixelshuffle method as an upsampling operation for our model. Finally, we can get the high-resolution images I_{SR} generated by generator network (G), and I_{SR} is represented as follows:

$$I_{\mathrm{SR}} = \mathrm{G}(I_{\mathrm{LR}}, \theta_{\mathrm{G}})$$

where $\theta_{\mathrm{G}} = W_{1-N}, b_{1-N}$ refers to the weights and biases of the N-layer generator, respectively.

2) Discriminator Network: These probabilities are calculated as follows:

$$P_{\mathrm{SR}}, P_{\mathrm{HR}} = D(I_{\mathrm{SR}}, I_{\mathrm{HR}}, \theta_D),$$

where $\theta_D = \{W_{1-M}, b_{1-M}$ refers to the weights and biases of the M-layer discriminator, respectively. We set the real label images I_{HR} and the generated images I_{SR} by the generator as inputs to the discriminator. Then, we set the category of I_{SR} to zero and the category of I_{HR} to one. Finally, our goal is to make the value of P_{SR} closer to zero and make the value of P_{HR} closer to one. By doing so, we can make the generated images I_{SR} closer to the label images I_{HR}.

C. Loss Function

Our simplified perceptual loss contains content loss and adversarial loss. We discard the regularization loss in SRCNN and define corresponding weights for the other two losses.

1) Content Loss: Instead of minimizing the mean square error, we aim to improve the perceptual similarity between the output images I_{SR} and the label images I_{HR}. Then, we denote the content loss as the Euclidean distance between the output I_{SR} and the label I_{HR}. This loss is caused by the generator, and based on the pretrained VGG network. Finally, we define this loss as follows:

$$T_{\mathrm{CL}(i,j)} = \frac{1}{M_{i,j}N_{i,j}} \sum_{x=1}^{M_{i,j}} \sum_{y=1}^{N_{i,j}} (\beta_{i,j}(I_{\mathrm{SR}})_{x,y} - \beta_{i,j}(G(I_{\mathrm{LR}})_{x,y})^2$$

where $M_{i,j}$ and $N_{i,j}$ denote the dimensions of the generated features in the generator in the VGG network, respectively. $\beta_{i,j}$ refers to the features produced by the jth Conv layer before the ith maxpooling within the VGG network.

2) Adversarial Loss: The loss encourages the images generated by our generator to fool the discriminator network, making the final generated images closer to the truth images. We apply nonsaturating loss in the discriminator, which provides a stronger gradient early in model training. Additionally, this loss is based on the probability of the discriminator on all input images, and we denote this loss as follows:

$$T_{\mathrm{AL}} = \sum_{l=1}^{L} -\log D(G(I_{\mathrm{LR}})).$$

Our perceptual loss T_{PL} is defined as follows:

$$T_{\mathrm{PL}} = \mu \cdot T_{\mathrm{CL}} + \delta \cdot T_{\mathrm{AL}}$$

where μ and δ are the balance coefficients separately. Furthermore, Adam optimizer is used to update model parameters in the generator and discriminator network.

III. Experiments

We compare DeepDT with five classic algorithms: Bilinear, SRCNN, LapSRN, SRGAN, and ESRGAN. Also, we compare our algorithms with two baseline models that are trained using MSE loss (DeepDT I), and the deconvolution upsampling layer (DeepDT II). Furthermore, we evaluate these models using six climate metrics, such as prediction omission (PO), false alarm ratio (FAR), and threat score (TS_i, $i = 1, 2, 3, 4$). Besides, we optimize the DeepDT network with a simplified perceptual loss and update model parameters with Adam optimizer. We set the learning rate to 10^{-6} for all layers in the network, and set another learning rate to 10^{-4} for the gradient descent. Additionally, the batch size of 32 and the epoch of 300 are set for our network to train. Furthermore, we train all networks on the Titan XP GPU and it greatly accelerates the training process. Note that our model input image has three channels and the final generated image only has one output channel.

A. Climate Data Set Description

According to the range of the precipitation peak, we select 2400 climate images as the NCNI data set which accounts for the same proportion in the 2017 precipitation observation data. For example, we divide four precipitation peak levels: drizzle ($\geq$1 mm), moderate rain ($\geq$20 mm), heavy rain ($\geq$50 mm), and rainstorm ($\geq$100 mm). Also, the NCNI data set records various climate information in central China, and each climate image records the weather information of a certain area within 10 min.

We use the NCNI benchmark data set for training, validating, and testing. Among them, we select 2000 climate images as our train data. In the model training process, we use random cropping operation for input images as a data augmentation method. Also, we use 200 images to validate the validity of our models and use 200 images to evaluate all networks in each climate metric. Meanwhile, we set the downscaling size to $\times$5 and $\times$10 km. Note that the pixels on each climate image record the precipitation value of the corresponding latitude and longitude coordinate points in central China at a certain moment.

TABLE I

QUALITATIVE COMPARISONS OF EIGHT ALGORITHMS IN TERMS OF SIX METRICS ON THE NCNI DATA SET. THE DOWNSCALING FACTOR IS ×10 km

Algorithm	PO	FAR	TS			
Bilinear	0.090	0.155	0.698	0.502	0.356	0.475
SRCNN	0.061	0.097	0.751	0.727	0.610	0.646
LapSRN	0.057	0.108	0.766	0.705	0.583	0.657
SRGAN	0.052	0.066	0.843	0.763	0.663	0.740
ESRGAN	0.051	**0.057**	0.851	0.764	0.665	0.739
DeepDT	**0.042**	0.058	**0.858**	**0.773**	**0.702**	**0.787**
Ablation Model						
DeepDT I	0.054	0.065	0.841	0.759	0.661	0.742
DeepDT II	0.051	0.064	0.845	0.768	0.664	0.751

TABLE II

QUALITATIVE COMPARISONS OF EIGHT METHODS IN TERMS OF SIX METRICS THE DOWNSCALING FACTOR IS ×5 km. THESE RESULTS ARE GENERATED BY AVERAGING THE VALIDATION DATA SET

Algorithm	PO	FAR	TS			
Bilinear	0.071	0.101	0.715	0.538	0.430	0.427
SRCNN	0.049	0.072	0.791	0.679	0.598	0.578
LapSRN	0.048	0.073	0.806	0.671	0.601	0.587
SRGAN	0.041	0.049	0.880	0.798	0.701	0.706
ESRGAN	0.042	0.042	0.891	0.808	0.705	0.704
DeepDT	**0.031**	**0.037**	**0.907**	**0.820**	**0.749**	**0.760**
Ablation Model						
DeepDT I	0.044	0.051	0.881	0.799	0.703	0.708
DeepDT II	0.041	0.046	0.893	0.801	0.711	0.716

B. Evaluation Metrics

To compare the performance of existing models with DeepDT on precipitation prediction, six climate evaluation metrics are presented: PO, FAR, and $\mathrm{TS}_i(i = 1, 2, 3, 4)$. According to the amount of precipitation, we divide four precipitation levels: drizzle, moderate rain, heavy rain, and rainstorm. These four levels correspond to four values of the TS score. Finally, these six indicators are calculated as follows:

$$\mathrm{PO} = \frac{\mathrm{NC}}{\mathrm{NA}+\mathrm{NC}} \times 100\%$$
$$\mathrm{FAR} = \frac{\mathrm{NB}}{\mathrm{NA}+\mathrm{NB}} \times 100\%$$
$$\mathrm{TS}_i = \frac{\sum_{\mathrm{day}=1}^{\mathrm{day}=n_i} \mathrm{NA}}{\sum_{\mathrm{day}=1}^{\mathrm{day}=n_i} (\mathrm{NA}+\mathrm{NB}+\mathrm{NC})} \times 100\%.$$

Among them, NA indicates the number of meteorological observatories predicted to be rainy in the weather forecast, NB refers to the number of outliers predicted by meteorological observation stations, NC indicates the number of climate observatory that do not predict the weather case in certain areas, and n_i refers to the number of days observed by the meteorological observatory for each precipitation level. Furthermore, the larger the TS scores are, the more accurate the forecast of precipitation levels. Also, the smaller the PO and FAR values are, the more complete precipitation forecast is.

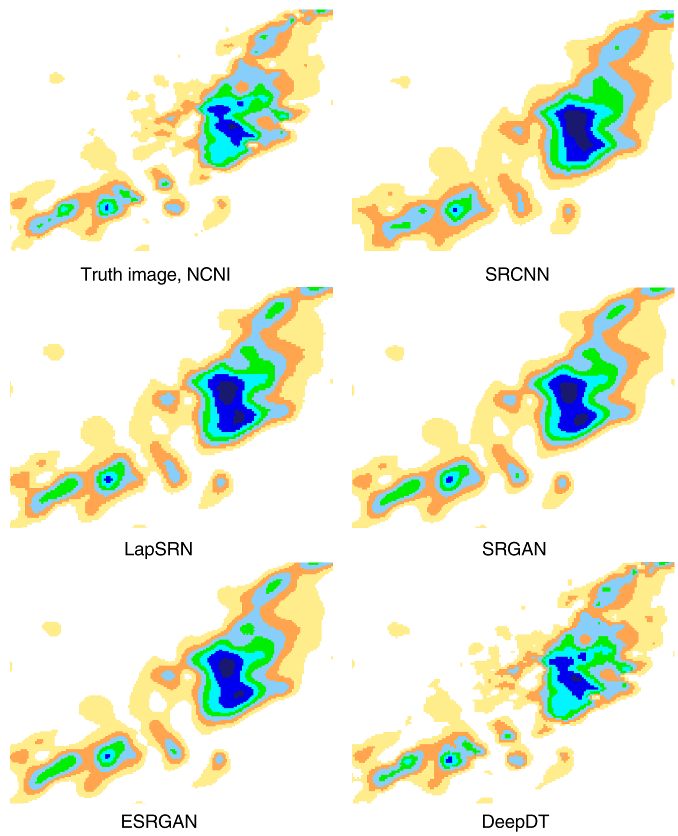

Fig. 5. Qualitative comparisons of the darker regions on the SRCNN, LapSRN, SRGAN, ESRGAN, and DeepDT. The scale factor is ×10 km.

C. Results and Evaluations

We compare all networks with the downscaling scale are ×5 and ×10 km on the NCNI data set, as shown in Tables I and II. Among them, when compared with the traditional super-resolution methods (Bilinear, SRCNN, and LapSRN), all GAN-based models (SRGAN, ESRGAN, and DeepDT) perform the best in each indicator with all downscaling scale. This indicates the great effectiveness of the GAN algorithms over the CNN-based models. When compared with the other two GAN-based networks (SRGAN and ESRGAN), our model still has further improvements in most metrics. Specifically, DeepDT achieves the most significant performance on the TS_3 and TS_4 scores. This shows that our algorithm outperforms in predicting heavy rain and rainstorm. All climate indexes show improved appearances for our DeepDT over these two ablation models (DeepDT I and DeppDT II), which indicates that our model uses the simplified perceptual loss, instead of MSE loss, dose further improves the performance of the model in weather forecasting. Additionally, the performance of our network will not be significantly improved compared to other models as the downscaling scale becomes smaller. We infer that the main reason is that the precipitation area in the climate image is small. When the scale factor is smaller, the downscaling technologies will not insert enough predicted values in the low-resolution image. This does not show the significant advantages of our model.

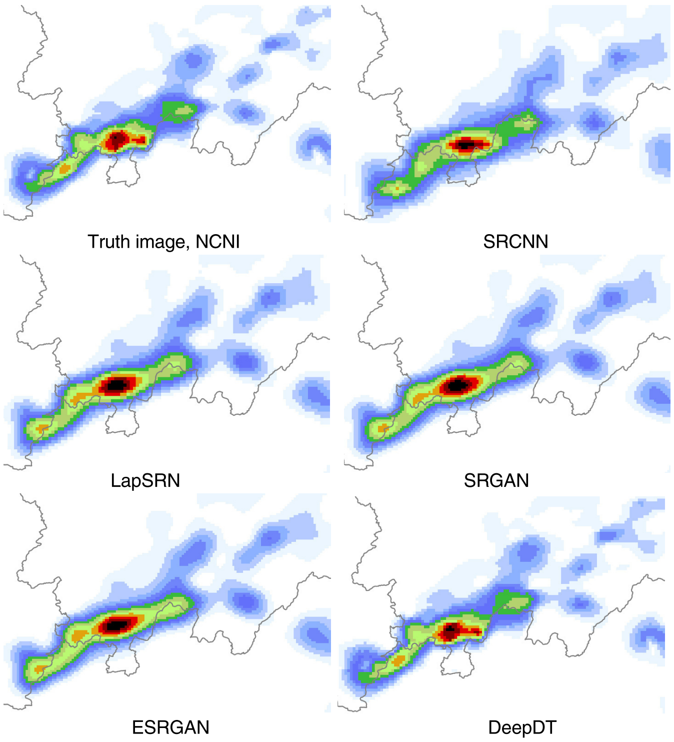

Fig. 6. Qualitative comparisons of the darker regions in the final output climate images. The scale factor is $\times 5$ km.

Fig. 5 shows qualitative comparisons of the darker regions in the generated climate image with the scale factor is, $\times 10$ km and we show the $\times 5$ km results on the NCNI data set in Fig. 6. Note that the darker the color is in the high-resolution climate image, the greater the amount of precipitation is in the area. It is obvious that some unpleasant noises appear in the images generated by the other four models, and it is easier to generate more precipitation values than the standard case. In contrast, the climate images produced by our DeepDT are closer to the truth images. Thus, we deduce that our method can effectively eliminate unpleasant artifacts in dark areas. Also, this shows that our proposed model achieves better performance in recovering details in the final generated images.

IV. CONCLUSION

In this letter, we propose an efficient GAN algorithm for high-resolution climate prediction. Our DeepDT keeps the high-level architectural design of SRGAN and applies the dense residual learning strategy to exploit feature maps. Also, we optimize the DeepDT using a simplified perceptual loss, which can recover more high-frequency content in the high-resolution images. Besides, we use a CNN method to fuse multiple related climate factors to generate climate images and construct a high-quality climate data set for precipitation prediction. Experimental results also illustrate that our method can generate high-quality climate images and outperforms most super-resolution models in climate prediction.

REFERENCES

[1] T. Vandal, E. Kodra, S. Ganguly, A. Michaelis, R. Nemani, and A. R. Ganguly, Eds., “Generating high resolution climate change projections through single image super-resolution: An abridged version,” in *Proc. Int. Joint Conf. Artif. Intell. Org.*, 2018, pp. 1–5.

[2] T. Vandal, E. Kodra, S. Ganguly, A. Michaelis, R. Nemani, and A. R. Ganguly, Eds., “DeepSD: Generating high resolution climate change projections through single image super-resolution,” in *Proc. 23rd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2017, pp. 1663–1672.

[3] M. Hessami, P. Gachon, T. B. M. J. Ouarda, and A. St-Hilaire, “Automated regression-based statistical downscaling tool,” *Environ. Model. Softw.*, vol. 23, no. 6, pp. 813–834, Jun. 2008.

[4] J. Yang, J. Wright, T. S. Huang, and M. YJItoip, “Image super-resolution via sparse representation,” *IEEE Trans. Image Process.*, vol. 19, no. 11, pp. 2861–2873, Nov. 2010.

[5] C. Dong, C. C. Loy, K. He, and X. Tang, Eds., “Learning a deep convolutional network for image super-resolution,” in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 184–199.

[6] C. Dong, C. C. Loy, K. He, and X. Tang, “Image super-resolution using deep convolutional networks,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2015.

[7] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, Eds., “Deep Laplacian pyramid networks for fast and accurate super-resolution,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 624–632.

[8] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, Eds., “Residual dense network for image super-resolution,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

[9] C. Ledig, L. Theis, F. Huszár, J. Caballero, A. Cunningham, and A. Acosta, Eds., “Photo-realistic single image super-resolution using a generative adversarial network,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4681–4690.

[10] A. Odena, V. Dumoulin, and C. Olah, “Deconvolution and checkerboard artifacts,” *Distill*, vol. 1, no. 10, p. e3, Oct. 2016.

[11] J. Johnson, A. Alahi, and L. Fei-Fei, “Perceptual losses for real-time style transfer and super-resolution,” 2016, *arXiv:1603.08155*. [Online]. Available: http://arxiv.org/abs/1603.08155

[12] X. Wang, K. Yu, C. Dong, and C. C. Loy, Eds., “Recovering realistic texture in image super-resolution by deep spatial feature transform,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 606–615.

[13] X. Wang, K. Yu, S. Wu, J. Gu, Y. Liu, and C. Dong, Eds., “ESRGAN: Enhanced super-resolution generative adversarial networks,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 1–16.

[14] M. M. Sheikholeslami, S. Nadi, A. A. Naeini, and P. Ghamisi, “An efficient deep unsupervised superresolution model for remote sensing images,” *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1937–1945, 2020.

[15] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, Eds., “Densely connected convolutional networks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.